

Project Vision & Requirements Specification

Project: Automated Employee Payroll Pipeline (AEPP)

Target: Nippon Toyota Internship Practical Evaluation (Task 1)

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Core Objective

To build a highly scalable, fast, and secure automated pipeline that ingests employee payroll data, generates professional salary slips, and seamlessly dispatches them to employees. The system must operate at an enterprise standard, capable of scaling from 50 to 1,000+ employees smoothly while maintaining absolute data security and zero operational exceptions.

System Components & Strict Expectations

1. Frontend Interface

What to Implement: An interactive web portal for administrators to upload payroll files and track the live progress of the automation pipeline.

Technology: Next.js.

Expectations: The interface must be a modern, sleek, enterprise-level UI/UX. It must not look like a typical, thrown-together "vibe-coded" template. Pages must be highly consistent, incredibly readable, and intuitively designed for professional administrative use.

2. Backend API Core

What to Implement: The central nervous system of the application that receives uploads, orchestrates the data pipeline, and manages state.

Technology: FastAPI.

Expectations: The backend must be designed for smooth scalability. It must feature robust security measures to handle external attacks seamlessly. It needs to be blazingly fast, highly responsive, and feature bulletproof error handling so that edge cases never disrupt the user experience.

3. Task Queue & Asynchronous Processing

What to Implement: A background processing engine to handle the heavy lifting of generating documents and sending emails independently from the main server.

Technology: Redis.

Expectations: This component must showcase the ability to write highly efficient, smart algorithms. It must intelligently handle missed or failed emails and employ custom or built-in logic to ensure the absolute maximum number of emails are delivered without exceptions. The architecture here must explicitly demonstrate high scalability to the recruiters.

4. Database & State Management

What to Implement: The permanent ledger tracking employee details and the real-time status of every single generated document and dispatched email.

Technology: PostgreSQL.

Expectations: The database operations must be exceptionally fast, highly scalable, and completely secure, aligning with professional data engineering standards to maintain perfect data integrity throughout the pipeline.

5. Document Generation Engine

What to Implement: A system that translates raw financial data into formatted, mathematical PDF documents.

Technology: Playwright (HTML-to-PDF).

Expectations: The output cannot be plain or generic. Every generated salary slip must be exceptionally well-styled, thoroughly professional, and highly readable for the end-user receiving it.

6. Email Dispatch System

What to Implement: The delivery mechanism that physically routes the generated PDFs to the respective employee inboxes.

Technology: Transactional Email API (Resend or SendGrid).

Expectations: The system must efficiently manage bulk dispatches without triggering spam filters or hitting rate limits, ensuring reliable and secure delivery to the employees.

7. Redundant Cloud Storage

What to Implement: A secure, transient holding area for the generated PDF files before and during the email dispatch process.

Technology: AWS S3.

Expectations: This storage must be impenetrable against attacks, treating the generated documents as highly sensitive financial information. It must act as a reliable redundancy layer—if an employee does not receive their slip, the document is safely stored for backup retrieval. To align with strict corporate compliance, the system must automatically delete these files after a designated period.